



SHADOWX

ShadowX User Manual

ICT3215 Digital Forensics PA-G 35	
Wu Yen Hao	2301919@sit.singaporetech.edu.sg
Lim Guo Dong	2301766@sit.singaporetech.edu.sg
Lang Jun Feng	2200566@sit.singaporetech.edu.sg
Tay Hao Xiang Ryan	2301851@sit.singaporetech.edu.sg
Lim Tze Kit	2301861@sit.singaporetech.edu.sg

Table of Content

- Introduction..... 3**
 - Installation and Setup.....3
- Using ShadowX..... 4**
 - Uploading of data.mdb file.....4
 - Starting the Monero service..... 4
 - Visualization..... 5
 - Block analysis lookup..... 7
 - Get block by height segment..... 7
 - Get transaction segment..... 8
 - Get outs segment..... 9
 - Age Based Bayesian Analysis..... 10
- Terminating ShadowX..... 13**
 - Terminating Monero Service..... 13
 - Terminating ShadowX..... 13

Introduction

Welcome to ShadowX, A Monero Analysis web tool

Currently, ShadowX is developed under the Monero Stagenet environment and in offline mode. The Monero stagenet service will be hosted at the device port 38081.

Installation and Setup

As this tool is currently only developed for usage in Kali Linux and suitable only in Linux environment, please take note of the following steps apply only in this environment

To install this tool, follow this step:

1. Clone the repository

```
git clone https://github.com/echoezz/ShadowX.git
```

2. Place the project directory ShadowX at your home directory of your Kali VM (Eg. /home/kali/ShadowX)
3. As github does not publish empty directories, you will need to create 3 folders inside the ShadowX folder. (Follow the naming convention)
 - Firstly, create the folder `uploadedFiles`
 - Then inside the folder `uploadedFiles`, create the folder `stagenet`
 - Lastly, inside the folder `stagenet`, create the folder `lmdb`

Your folder structure should look like this:

```
ShadowX/
├── templates/
├── uploadedFiles/
│   ├── stagenet
│   │   └── lmdb
├── static/
├── analysis/
├── monero-x86_64-linux-gnu-v0.18.4.4/
├── app.py
└── img/
```

4. Navigate to the folder `monero-x86_64-linux-gnu-v0.18.4.4` and execute the command to give `monerod` executable permission.

```
chmod +x monerod
```

5. To start the web application, navigate to the ShadowX directory folder and run the following command in the terminal

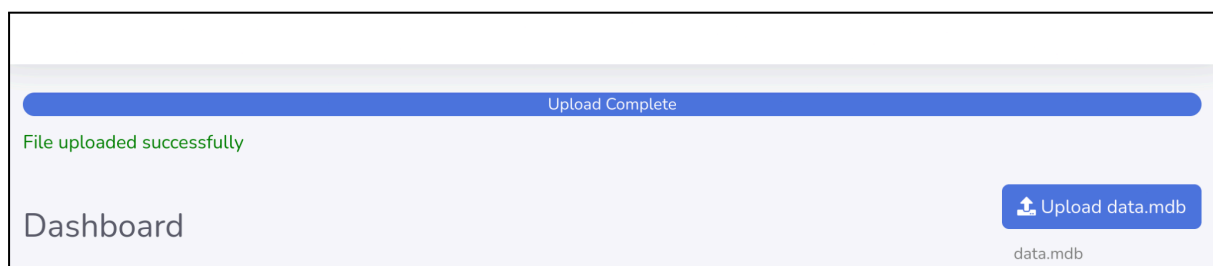
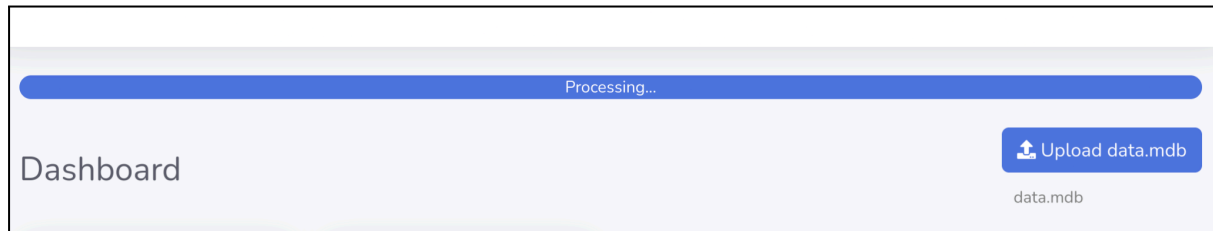
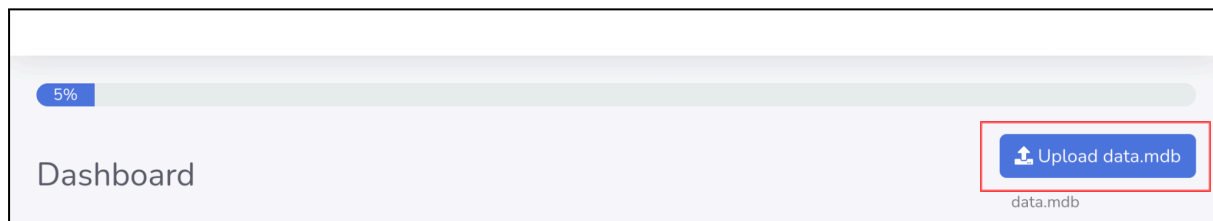
```
python3 app.py
```

Open your browser and visit the page at localhost:5000 to start using ShadowX

Using ShadowX

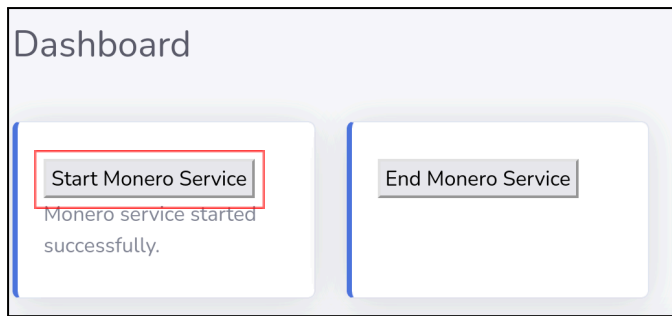
Uploading of data.mdb file

At the **index** page, click the button “Upload data.mdb” button to upload the data.mdb file you have and wait for the file to finish uploading



Starting the Monero service

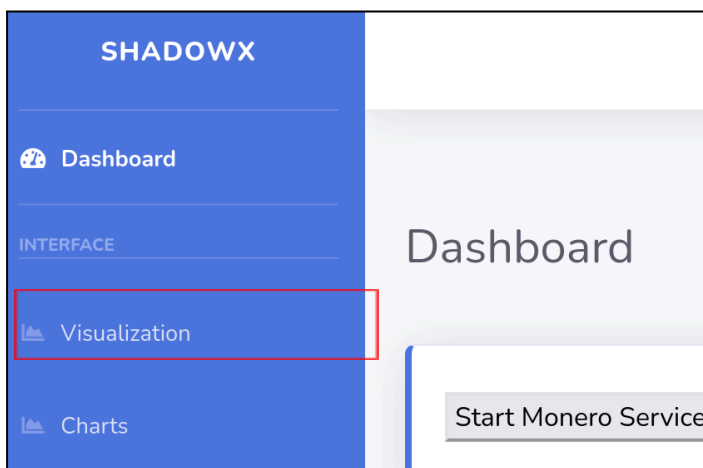
At the index page, after uploading the data.mdb file successfully, start the Monero service by clicking the button “Start Monero Service”



Once started, we can begin to perform the analysis

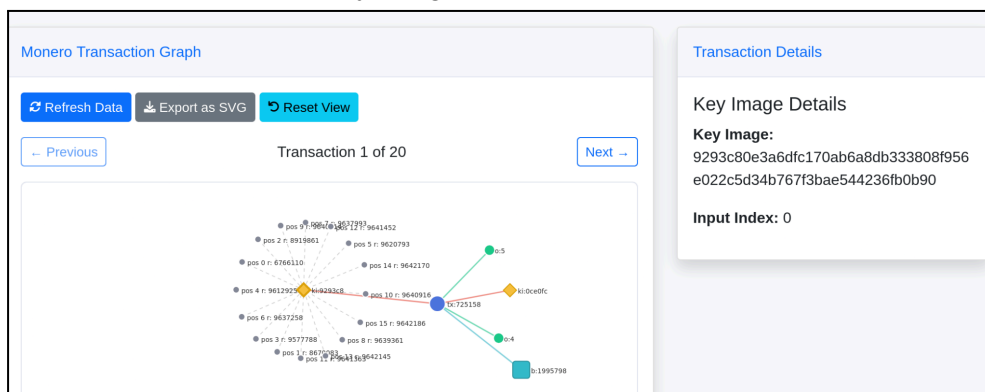
Visualization

Users can navigate to the visualization page by pressing on the visualization tab at the side. At the visualization page, once the Monero service is started, users can view the different transactions within the data.mdb file (For this project, we only scale up to 20 transactions to reduce overhead)



Users can interact with the nodes by clicking on it, which will return the respective information as such:

- Yellow nodes for Key Image Details



- Blue nodes for Transaction details

Monero Transaction Graph

Refresh Data
Export as SVG
Reset View

Previous
Transaction 1 of 20
Next

Transaction Details

Transaction Details

Hash:
725158b15ada7cdcef93a8e77efc97a662f9bd184d88848b9c3f44cc433f08c5

Block Height: 1995798

Fee: 0.000059430000 XMR

Inputs: 2

Outputs: 2

Version: 2

- Green node for Output details

Monero Transaction Graph

Refresh Data
Export as SVG
Reset View

Previous
Transaction 1 of 20
Next

Output Details

Output Details

Index: 5

Amount: 0.000000000000 XMR

This output can be spent in future transactions.

- Square node (turquoise) for Block Details

Monero Transaction Graph

Refresh Data
Export as SVG
Reset View

Previous
Transaction 1 of 20
Next

Transaction Details

Block Details

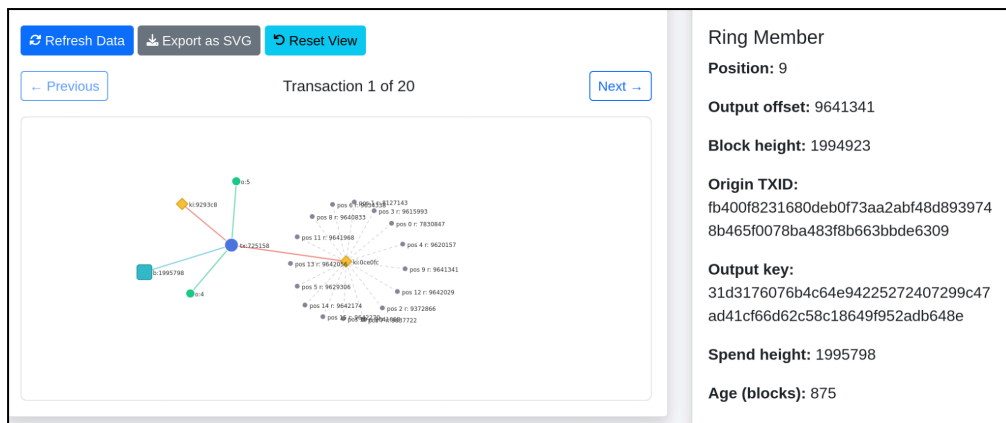
Height: 1995798

Hash:
4cb41dd54bb3c61f9caa369ed5dcd6e2d70477e93a18933b04831d3e0768276f

Timestamp: 11/21/2025, 10:48:16 AM

Transaction Count: 1

- Grey node for Ring Member Details



Users can also choose to export the visualization if they wish to do so via the “Export as SVG” button

Monero Transaction Graph



Block analysis lookup

If users require more information, they can perform block analysis by looking up values they see from the visualization, after interacting with the nodes.

Get block by height segment

Users can input a block height of their interest and perform a search

Block analysis

Enter Block Height:

They will then be able to gain more information from the block height as follows for analysis purposes:

Block Information

Block Height: 1994923

Block Hash:
23652f8eb2f8125473cc397217c33c0c9bbf43b511dae19c80513a7b0e9ddf35

Timestamp: 11/20/2025, 7:19:55 AM

Size: 3380

Difficulty: 342859

Cumulative Difficulty: 548610036416

Major/Minor Version: 16 / 16

Nonce: 2372700180

Miner Reward: 0.808288267326 XMR

Get transaction segment

Users can also perform a get_transaction lookup by supplying the transactional hash they retrieved from the visualization page earlier after their interaction with the nodes

Get transactions

Enter txs hash

ad90091169f01d74030b8df

Search

Get Transaction output

They will then be able to retrieve information such as the key image, key offsets and absolute indices

Get Transaction output

Version: 2

Unlock Time: 0

Inputs (Key Offsets):

Input 1:

Amount: 0

Key Image: b47a8e3ac736024919da793ac912485a223fadee517c79fe6426f2a4afdd262c

Key Offsets:

Key Offsets:

0: 9556354
1: 60413
2: 12062
3: 9355
4: 315
5: 697

Absolute Indices:

Index 0: 9556354
Index 1: 9616767
Index 2: 9628829
Index 3: 9638184
Index 4: 9638499
Index 5: 9639106

Get outs segment

If they wish to make use of the `get_outs` function, users can take the 16 absolute indices and fill up accordingly to the index position:

Get Outputs

Enter up to 16 outputs (amount and index pairs)

Amount	Index	Amount	Index
0	9378334	0	9640622
0	9591516	0	9640727
0	9616751	0	9641866

Amount	Index	Amount	Index
0	9378334	0	9640622
0	9591516	0	9640727
0	9616751	0	9641866
0	9635247	0	9641959
0	9635399	0	9642008
0	9637897	0	9642064
0	9638800	0	9642099
0	9639535	0	9642145

Once they submit via the “Get Outputs” button, they can retrieve information such as Block height, Key, Mask, Transaction ID:

Output Results

Retrieved Outputs:

Output 1:

Height: 1773130

Key: fd628c57e95b0bc702afc780709157f3395ee1c318daad431b408ab89dc0582e

Mask: dbca8d74478047294b6d1703a46550204cfd5911ee10be97186bfd1d9caee9f

Unlocked: true

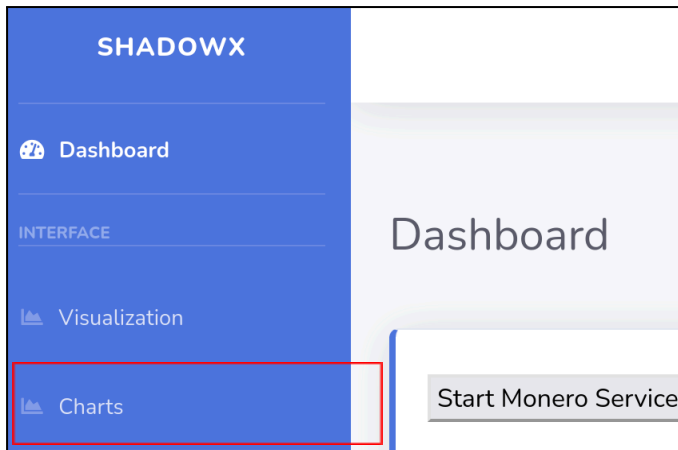
Transaction ID: 2917ac986a30e0709cd79283b215e426eff5262eae2e67036dbb3760411a6b42

Output 2:

Height: 1955141

Age Based Bayesian Analysis

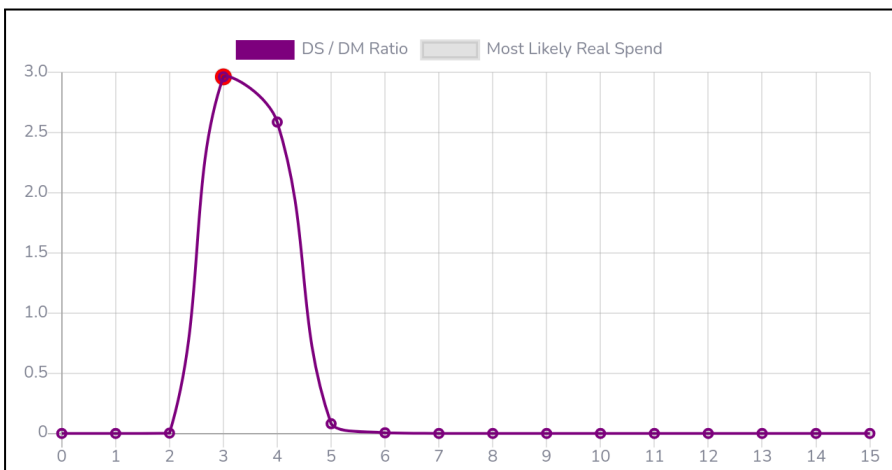
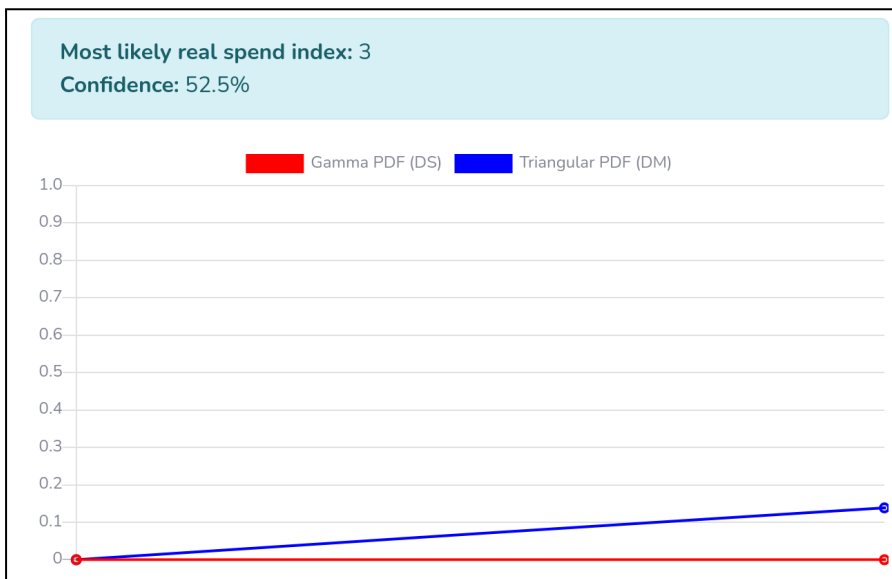
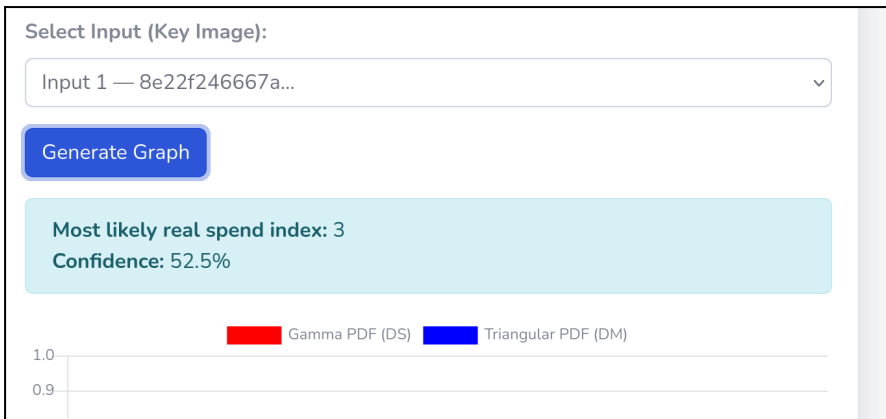
If the user wished to perform Age Based Bayesian Analysis on attempt to analyse the decoys previously seen from the visualization page to determine the likely real spent of a transaction, they can head over to the charts page with a transactional hash taken from the visualization page



They can perform a search with the transactional hash and click on the “Load Inputs” button

If the transactional hash has multiple key images, it will show up accordingly for the user to choose from. They can then generate a graph by clicking on the “Generate Graph” button

After clicking on the “Generate Graph” button, they can view the following analysis that makes use of Age Based Bayesian Analysis to try and locate the likely real spent of a transaction.



Terminating ShadowX

Terminating Monero Service

Once a user is done using ShadowX, they can proceed to the index page and click on the “End Monero Service” button to terminate ShadowX and the Monero service properly.



Users can also run the following command on their terminal in Kali Linux to ensure the Monero service is shutdown successfully.

```
ps aux | grep monerod
```

Terminating ShadowX

To terminate ShadowX, at the terminal where app.py was launched, execute Ctrl C to end the application.

End of user manual